CHRIST (Deemed to be University), Bangalore – 560 029
Department of Computer Science

END SEMESTER EXAMINATION – MAY 2024
UG IV Semester

Programme Name: BCA **Max. Marks: 100**

Course Name: Software Engineering **Time: 3 Hrs**

Course Code: BCA434

General Instructions

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / Course title & number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place in the hall.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer booklet. No sheets should be detached from the answer booklet.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer booklet.

Course Outcomes (COs): The students will be able to

CO1. To learn the basic principles of software engineering.

CO2. Understand the various process models and the stages in the software life cycle.

CO3. Design and apply different elements of the user interface using golden rules.

CO4. Analyse and apply different software testing approaches to improve the quality of software projects to cater the needs of society.

SECTION A

Answer ALL the Questions **10 X 2 = 20**

Q. No	Questions	CO	RBT
1	List any two System softwares.	CO 1	L1
2	"Software does not wear out". Comment.	CO	L3

		1	
3	Discover the list of things a requirement engineer can find during an inception task.	CO 2	L2
4	Generate a 0-level DFD for a specific application domain.	CO 3	L6
5	Justify the importance of functional independence among the modules for a good software design.	CO 3	L4
6	Verify the importance of refactoring in building a good software design.	CO 4	L5
7	Summarize the important tasks of SQA.	CO 4	L2
8	SQA actions are performed to achieve pragmatic goals. Mention any four of the important goals.	CO 4	L1
9	Infer on Validation testing.	CO 4	L3
10	Sketch risk table for a gaming project	CO 4	L6

SECTION B

Answer any <u>FIVE</u> Questions **5 X 6 = 30**

Q. No	Questions	CO	RBT
11	Elucidate concurrent and evolutionary process models.	CO1	L1
12	Evaluate the working principle and functionality of Object-	CO1	L5

	Oriented Design with a suitable diagram.		
13	Describe Class based modelling and Class Responsibility Collaborator modelling in software requirements.	CO2	L2
14	Illustrate the process of formal and informal technical reviews in detail.	CO2	L4
15	Distinguish the six sigma steps adopted for software engineering.	CO1	L3
16	Create various black box test cases involved in testing a Ticket Booking System.	CO4	L6
17	Elaborate on project scheduling with a real time instance.	CO4	L1

SECTION C

Answer any <u>FIVE</u> Questions **5 X 10 = 50**

Q.No	Questions	CO	RBT
18	Discuss Incremental process model with its applications and limitations	CO1	L3
19	Examine various steps involved in "establishing the groundwork" as part of requirement engineering.	CO2	L1
20	Depict the high-level functionality of any online Payment System with a Use Case diagram.	CO3	L6
21	Evaluate the McCall’s Quality factors and targeted quality factors for Software Quality Assurance.	CO2	L5

22	Analyse the steps to Develop Software Quality Assurance Plan for Smart Home Automation System	CO4	L4
23	Explain the necessary steps involved in Software Testing starting from Code to System Engineering.	CO4	L1
24	Discuss about Software project estimation and the types of decomposition techniques. Elaborate on any three techniques in detail.	CO4	L2

Revised Bloom's Taxonomy (RBT) Levels:		
L1 – Remembering	L2 – Understanding	L3 – Applying
L4 – Analyzing	L5 – Evaluating	L6 – Creating